Pairs Trading with Honest Cost Modeling: A Pre-Registered Case Study of How Transaction Costs Erode a Small Statistical-Arbitrage Edge

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Abstract

This paper measures how realistic transaction costs change the observed performance of a simple cointegration-based pairs-trading strategy, using a design in which every research choice (universe, sample split, selection rule, signal thresholds, execution timing, accounting rules and cost model) was recorded before any price data was downloaded. From all 28 pairs of eight US oil and gas producers, the pre-registered rule selects EOG Resources / Diamondback Energy (EOG/FANG), with a raw Engle-Granger p-value of 0.0051. After Holm correction for 28 tests no pair is significant (adjusted p = 0.142), and cointegration is not detected in the out-of-sample period (p = 0.223). Traded out of sample over 2021–2025 with $100,000 of gross notional per trade, the strategy earns +3.25% gross (Sharpe 0.08) and +0.72% net of costs (Sharpe 0.02) on the identical 20 trades, so costs consume 78% of the gross profit. The mechanism is turnover: each round trip trades about $202,000 of notional, so 5 bps per side of half-spread and slippage alone costs about $101 per trade, against an average gross gain of $163. The net profit disappears at 1.28 times the base-case costs. Neither the gross nor the net return is statistically distinguishable from zero. The contribution is methodological rather than a new anomaly: a reproducible, audit-trailed template for separating gross from net performance and for reporting a multiple-testing failure instead of hiding it.

Keywords: pairs trading, statistical arbitrage, cointegration, transaction costs, backtesting, multiple testing, pre-registration, reproducibility

JEL classification: G11, G14, C12, C32

1. Introduction

Pairs trading is one of the oldest and most widely taught statistical arbitrage strategies. The idea is simple: find two assets whose prices share a long-run relationship, wait for the spread between them to move unusually far from its norm, then buy the relatively cheap asset, sell the relatively expensive one, and close the position when the spread reverts. Because the idea is so simple, it is also easy to backtest badly. Three problems appear again and again in informal backtests:

1. **Costs are ignored or added as an afterthought.** A strategy that trades frequently can look profitable gross of costs and lose money net of them.

2. **The pair is chosen by searching.** Testing many candidate pairs and keeping the best one makes the winner look more significant than it is.
3. **Future information leaks into the past.** Choosing the pair, fitting the hedge ratio or normalizing the signal on the same data that is then traded inflates performance.

This paper asks a narrow question: *how does incorporating realistic transaction costs affect the observed performance of a pairs-trading strategy?* It answers it for one pair, selected by a fixed rule, with the gross and net results computed from the same trades. The gross result was recorded and cryptographically hashed before any cost-model code existed, so it could not be adjusted after the net result was seen.

The finding is that a small gross edge (+$163 per trade, or 8.1 basis points of the notional each trade turns over) is mostly consumed by a modest cost model (about $127 per trade, or 6.3 basis points), leaving a net result that is positive but economically and statistically indistinguishable from zero. The pair itself does not survive correction for the 28 pairs that were searched.

None of this is a new empirical discovery. The sensitivity of pairs-trading profits to trading costs is well documented (Section 2). The contribution is a worked, fully disclosed example of doing the analysis honestly: every decision is recorded as a dated decision record (ADR) in the repository, every candidate pair is reported, the weak statistical evidence is stated up front, and the published numbers can be reproduced exactly from a raw data snapshot identified by its SHA-256 hash.

2. Related work

Gatev, Goetzmann and Rouwenhorst (2006) documented economically large historical excess returns to a simple distance-based pairs rule in US equities. Later work found that these profits declined over time and were sensitive to how trading costs are modelled (Do and Faff, 2010, 2012). Krauss (2017) reviews the main families of pairs-trading methods, including the distance, cointegration, time-series and stochastic-control approaches. The cointegration approach used here follows the two-step procedure of Engle and Granger (1987) as popularized for trading by Vidyamurthy (2004), with p-values from MacKinnon (2010).

More broadly, Novy-Marx and Velikov (2016) show that trading costs substantially reduce the net returns of many published return anomalies, especially high-turnover strategies. On the statistical side, Harvey, Liu and Zhu (2016) argue that the large number of strategies tested in finance requires much higher significance hurdles, and Bailey, Borwein, López de Prado and Zhu (2014) show how backtest overfitting produces strategies that fail out of sample. This paper applies the simplest version of that discipline, a Holm (1979) correction over every pair tested, and borrows the idea of pre-registration from the reproducibility literature (Nosek et al., 2018).

3. Research design

All parameters below were fixed in the repository's `DECISIONS.md` on 2026-09-23 before any price data was downloaded (ADR-001 to ADR-004, ADR-006 and ADR-007). Mechanical details that the original records left open were fixed in later ADRs, each recorded before the step it governs was run on real data (ADR-009 to ADR-011). No parameter was changed after any result was seen.

3.1 Data and universe

The candidate universe is eight US-listed large-cap oil and gas producers: XOM, CVX, COP, EOG, OXY, DVN, APA and FANG. Their revenues share a dominant common driver (crude oil and natural gas prices), which is the standard economic argument for why their prices might share a long-run stochastic trend. Daily adjusted and unadjusted closing prices come from Yahoo Finance via `yfinance` (version 1.7.0), from 2016-01-04 to 2025-12-31. Only dates on which all eight stocks trade are kept; none were dropped, giving 2,514 trading days. No values are filled, interpolated or repaired. The raw snapshot's SHA-256 hash is recorded in `data/dataset_manifest.json`.

The sample is split 50/50 by calendar year, as a rule rather than after inspecting the data:

- **Formation period, 2016–2020 (1,259 days):** cointegration testing, pair selection and hedge-ratio estimation.
- **Trading period, 2021–2025 (1,255 days):** every reported performance number.

3.2 Pair selection and multiple testing

Each of the $\binom{8}{2} = 28$ pairs is tested with the Engle-Granger two-step test on log adjusted closes over the formation period (`statsmodels` `coint`, constant term, lag length by AIC). The dependent variable is the alphabetically first ticker of each pair, an arbitrary rule fixed in advance; the reverse ordering is reported as a robustness check. The pre-registered selection rule is to trade the pair with the lowest raw p-value. Raw, Holm-adjusted and Bonferroni-adjusted p-values are reported for all 28 pairs (Appendix A).

A pre-registered *weak-relationship rule* required that, if the selected pair did not survive Holm correction at 5%, this be reported as a finding and raised before any backtest, and that the pair not be silently replaced.

3.3 Spread, signal and execution

The hedge ratio β and intercept α come from an OLS regression of $\log Y$ on a constant and $\log X$ over the formation period only, and are held fixed for the whole trading period. The traded spread is

$$s_t = \log Y_t - \beta \log X_t - \alpha,$$

and its z-score uses a trailing 60-day window that ends at, and includes, day t:

$$z_t = \frac{s_t - \bar{s}_{t-59:t}}{\hat{\sigma}_{t-59:t}}.$$

The strategy enters short-spread (short Y, long X) when $z_t \geq +2$ and long-spread when $z_t \leq -2$. It exits a short-spread position when $z_t \leq +0.5$ and a long-spread position when $z_t \geq -0.5$, which also covers the case where z overshoots through the exit band in one day (ADR-009). There is no stop-loss. A position decided at the close of day t is executed at the close of day $t + 1$. Any open position is closed at the final close of the trading period.

Each trade uses $100,000 of gross notional, split between the legs in the ratio $1 : \beta$, in whole shares computed from the unadjusted close at execution. Positions are not compounded. P&L uses dividend-adjusted closes, so the short leg pays dividends and the long leg receives them. Idle cash earns nothing. The Sharpe ratio is annualized with $\sqrt{252}$ over every trading-period day, including flat days, with a 0% risk-free rate (ADR-006, ADR-010).

3.4 Transaction cost model

Costs are charged per leg, per order, to traded notional:

Component	Base case	Basis
Half bid-ask spread	2 bps	Assumption for liquid US large caps
Slippage	3 bps	Assumption for execution near the close at small size
Commission	$0.005/share, $1 minimum per order	Published fixed-rate broker schedule
Short borrow	0.30% per year on short market value	General-collateral assumption

These are assumptions, not measurements: daily Yahoo data contains no quote information. Regulatory fees, market impact beyond slippage, borrow recalls and margin interest are not modelled. Dividends paid on the short leg are already inside gross P&L through the adjusted prices and are not charged a second time. The net result is also reported at 0×, 0.5×, 2× and 3× the base case, with every component, including the commission minimum and borrow rate, scaled together.

3.5 Bias controls and verification

Look-ahead. The pair and hedge ratio use formation data only; the z-score uses only past and current values; and execution lags the signal by one day. A unit test perturbs a single future price of either stock at several dates and checks that every earlier z-score and position is unchanged. Another test checks that trading-period prices cannot change pair selection.

Frozen gross result. The gross backtest was run and its output file hashed (SHA-256 `1acbfd35…df853`) before any cost code was written. The net run reads the 20 frozen trades rather than regenerating them, verifies the hash, rebuilds every trade's gross P&L to check it against the frozen values, and confirms that 0× costs reproduce the gross result exactly before applying costs (ADR-010, ADR-011). Two trades were also recomputed independently from raw prices and match to the cent.

Survivorship. Not corrected; see Section 6.

4. Results

4.1 Pair selection

Five of the 28 pairs have a raw p-value below 0.05, against about 1.4 expected by chance if no pair were cointegrated. The tests are not independent, since every pair shares the same oil-price factor, so this count is only a rough guide.

Rank	Pair (Y/X)	EG statistic	Raw p	Holm p	Hedge ratio β
1	**EOG/FANG**	−4.105	**0.0051**	**0.142**	0.742
2	FANG/OXY	−3.950	0.0085	0.228	0.701
3	APA/DVN	−3.484	0.0338	0.880	1.097
4	EOG/OXY	−3.436	0.0385	0.962	0.527
5	COP/CVX	−3.395	0.0429	1.000	1.446

Table 1. The five pairs with the lowest raw p-values, formation period 2016–2020. The 5% critical value is −3.341. All 28 pairs are in Appendix A.

The selected pair, EOG/FANG, is economically plausible: both are US shale oil producers with large Permian Basin operations and exposure to the same crude price, basin differentials and service-cost cycle. Statistically, however, it does not survive correction: its Holm-adjusted p-value is 0.142, and no pair is significant at 5% after correction. The weak-relationship rule was triggered. Following the pre-registered rules, the project continued with EOG/FANG and disclosed the failure prominently (ADR-008). Every result below should therefore be read as a test of a pair *found by searching 28 candidates*, not of a statistically established relationship.

The formation period is also unusual: it is dominated by the March 2020 crash, in which every ticker had its largest daily move.

4.2 Out-of-sample stability

Re-running the same Engle-Granger test on the trading period gives a statistic of −2.638 and p = 0.223, so cointegration is not detected out of sample. The hedge ratio itself is stable (0.742 in the formation period, 0.754 in the trading period). A supplementary, non-pre-registered ADF test on

the spread actually traded (with the fixed formation coefficients) gives p = 0.080: weak evidence of mean reversion, not significant at 5%.

4.3 Gross versus net performance

Metric (EOG/FANG, 2021–2025)	Gross of costs	Net of costs
Total P&L on $100,000	+$3,253	+$715
Total return	+3.25%	+0.72%
Annualized return	0.65%	0.14%
Annualized volatility	8.29%	8.29%
Sharpe ratio	0.08	0.02
Approximate t-statistic	0.18	0.04
Maximum drawdown	−11.35%	−12.28%
Win rate	60.0%	60.0%
Average P&L per trade	+$163	+$36
Round-trip trades	20	20 (identical)

Table 2. Gross and net results from the same 20 trades. The approximate t-statistic is the annualized Sharpe ratio times the square root of the number of years (five).

The strategy was in the market 44.7% of the time, held trades for 28.1 trading days on average, and turned over 8.1 times its capital per year. Costs total $2,538, equal to 78% of the gross P&L. The win rate does not change because no winning trade was small enough to be turned into a loss by its cost.

4.4 The mechanism: turnover against a thin edge

The cost drag is large even though the cost model is modest, because of how much notional each trade moves. A round trip enters and exits both legs, so a trade with $100,000 of gross notional turns over about $202,000. At 5 bps per side for half-spread plus slippage, that alone costs about $101. Commissions add about $9 and short borrow about $17 over an average 28-day hold, for about $127 per trade.

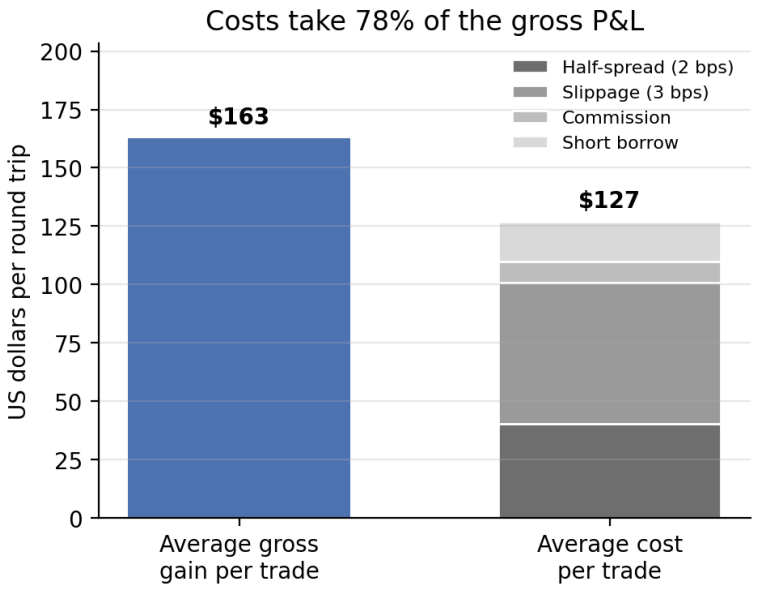


Figure 1. Average gross gain per trade compared with the average cost per trade, broken down by component.

The useful way to see the result is per unit traded. The strategy's average gross gain is 8.1 bps of the notional it trades; its cost is about 6.3 bps. The edge is only about 1.3 times the cost, which is why a moderate error in the cost assumptions is enough to reverse the sign of the result. Slippage (48% of costs) and half-spread (32%) dominate; commission (7%) and borrow (14%) are secondary.

4.5 Sensitivity to the cost assumption

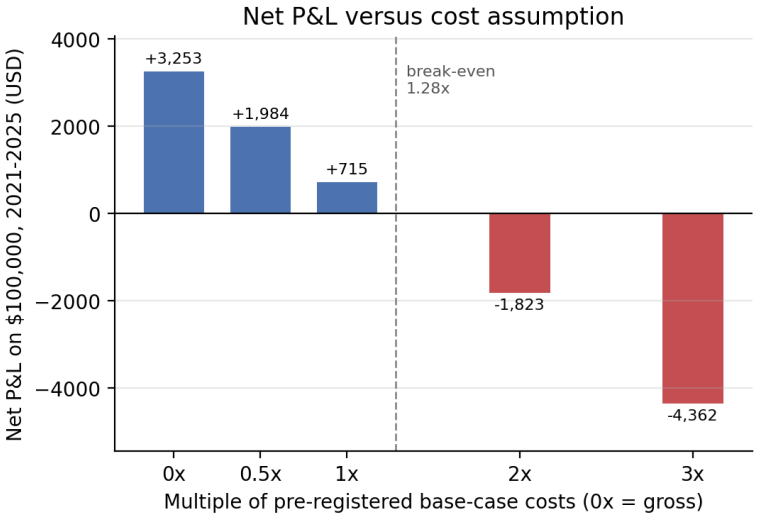


Figure 2. Net P&L at 0, 0.5, 1, 2 and 3 times the base-case costs, with the break-even multiple marked.

Cost multiplier	0× (gross)	0.5×	**1× (base)**	2×	3×
Total costs	\$0	\$1,269	\$2,538	\$5,077	\$7,615
Net P&L	+\$3,253	+\$1,984	**+\$715**	−\$1,823	−\$4,362
Sharpe ratio	0.08	0.05	0.02	−0.04	−0.11

Table 3. Pre-registered cost sensitivity.

Net P&L falls linearly in the cost multiplier and reaches zero at 1.28 times the base case. Because the base case is itself an assumption for liquid large caps, a real-world cost only 28% higher than assumed, which is well within the uncertainty of the estimate, eliminates the profit.

4.6 Trade-level behaviour

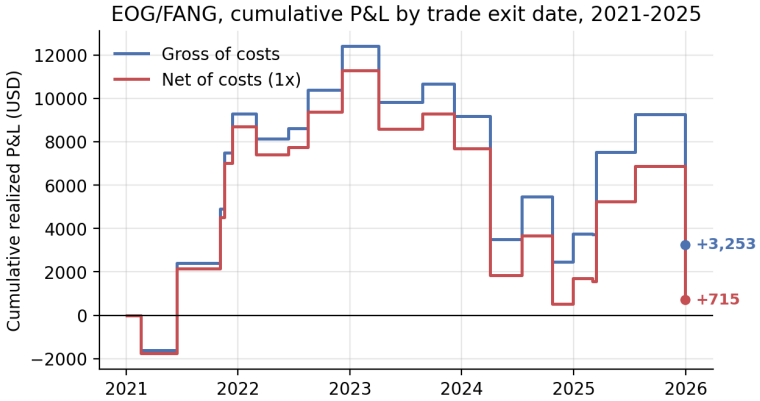


Figure 3. Cumulative realized P&L by trade exit date, gross and net of costs.

Figure 3 shows that the result is not a steady accumulation of small gains. Cumulative gross P&L peaked at about \$12,400 in December 2022 and was then eroded by a small number of large losses. The two largest losses (−\$5,674 and −\$6,014 gross) came from long-spread trades held 57 and 65 trading days while the spread kept diverging; the second was still open at the end of the sample and was closed by the forced final exit. A stop-loss would have limited these losses, but none was pre-registered, and adding one after seeing these trades would be fitting the rule to the result.

5. Discussion

The central lesson is about the ratio between edge and cost, not the level of either. A backtest that reports only gross performance hides this ratio entirely. Here, the gross result alone would already have been unimpressive (Sharpe 0.08, $t \approx 0.18$); the net result shows that even that small edge is mostly a payment to liquidity providers under plausible assumptions.

The second lesson is about search. Choosing the best of 28 correlated tests produced a raw p-value of 0.0051 that looks convincing in isolation and is not significant once the search is accounted for.

The same relationship was not detected in the following five years. A reader who saw only the selected pair and its raw p-value would be badly misled.

The third lesson is procedural. Freezing the gross result before writing cost code, reusing the identical trades for the net run, and verifying both with hashes makes the gross-versus-net comparison auditable. The design choices that constrain the analysis (no stop-loss, a single pair, fixed thresholds) are limitations, but they are also what make the result credible: nothing was tuned after the fact.

6. Limitations

- **Statistical power.** One pair and 20 trades over five years cannot establish or rule out a small edge. Neither the gross nor the net return is distinguishable from zero.
- **The pair was not statistically validated.** It failed Holm correction in the formation period and showed no cointegration out of sample.
- **Survivorship bias.** All eight tickers are listed today. Producers that were acquired or delisted during 2016–2025 (for example Anadarko, Noble Energy, Concho, Pioneer, Marathon Oil and Hess) are excluded, partly because free data for delisted tickers is unreliable. Choosing an energy universe also reflects general market knowledge.
- **Corporate actions.** Several firms completed large mergers during the sample, changing the businesses behind the tickers.
- **Assumed, not measured, costs.** Spread, slippage and borrow are assumptions. Market impact, regulatory fees, borrow recalls and margin interest are not modelled, and idle cash and short proceeds earn nothing.
- **No stop-loss and fixed parameters.** Thresholds (± 2.0 entry, 0.5 exit, 60-day window) are conventional, not tuned, and no sensitivity analysis over them was run.
- **Data.** Daily, free public data only. Yahoo revises adjusted history, so a fresh download may differ from the snapshot used here.
- **Generality.** A single pair in a single sector and period supports no claim about pairs trading in general.

7. Conclusion

Under a pre-registered design, a cointegration-based pairs trade in EOG/FANG earned a small, statistically weak gross profit over 2021–2025, and a modest, explicitly stated cost model removed 78% of it. The net result disappears if costs are 28% higher than assumed. The selected pair was the best of 28 searched candidates and did not survive correction for that search. The practical conclusion is that the gross edge per unit traded must be compared with the cost per unit traded before a pairs-trading backtest is taken seriously, and that the full search behind a selected pair must be disclosed.

A natural extension is to apply the same frozen-gross, pre-registered protocol to a much larger

universe of pairs across several sectors and periods, including delisted firms, with costs estimated from quote data rather than assumed.

Data and code availability

All code, tests, decision records and generated results are available at https://github.com/mateeniqbal3/pairs-trading-honest-costs under the MIT License, and each release is archived on Zenodo (the DOI is listed in the repository README). Raw prices are not redistributed. The exact inputs and outputs are identified by SHA-256:

- Raw price snapshot: `59106b32a3e642f557cc0c66578c386bf7c6eb0f48e9ecb9b017aa7e570823e2`
- Frozen gross results (`docs/results_gross.json`): `1acbfd35e8c899173a6e4cb355942cc5485bf01d5ac5d0bfc419d429746df853`
- Net results (`docs/results_net.json`): `06540672fd0013278661e23f0e78996ba763b836e732a5a8a23d60db802e55ca`

The full pipeline runs with `bash scripts/run_pipeline.sh`, and the figures in this paper are produced by `python paper/make_figures.py` from the committed result files only.

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Appendix A. Full pair-testing record (formation period 2016–2020)

Engle-Granger test on log adjusted closes, 1,259 observations per pair, 5% critical value −3.341. Holm and Bonferroni adjustments are over all 28 tests. "Reverse p" swaps Y and X and was not used for selection.

Rank	Pair (Y/X)	EG stat	Raw p	Holm p	Bonferroni p	β	Reverse p
1	EOG/FANG	−4.105	0.0051	0.1418	0.1418	0.7423	0.0036
2	FANG/OXY	−3.950	0.0085	0.2283	0.2367	0.7012	0.0491
3	APA/DVN	−3.484	0.0338	0.8795	0.9472	1.0970	0.0468
4	EOG/OXY	−3.436	0.0385	0.9619	1.0000	0.5267	0.0824
5	COP/CVX	−3.395	0.0429	1.0000	1.0000	1.4457	0.0081
6	OXY/XOM	−3.324	0.0515	1.0000	1.0000	2.4336	0.0432
7	FANG/XOM	−3.269	0.0591	1.0000	1.0000	1.7479	0.1663
8	CVX/XOM	−3.156	0.0776	1.0000	1.0000	0.3662	0.9613
9	CVX/OXY	−3.054	0.0980	1.0000	1.0000	0.1324	0.8973
10	CVX/EOG	−2.837	0.1543	1.0000	1.0000	0.3232	0.9126
11	CVX/FANG	−2.834	0.1550	1.0000	1.0000	0.2529	0.7540
12	DVN/FANG	−2.834	0.1552	1.0000	1.0000	1.0396	0.4614
13	CVX/DVN	−2.754	0.1806	1.0000	1.0000	0.1512	0.8579
14	APA/XOM	−2.558	0.2539	1.0000	1.0000	2.2048	0.4849
15	APA/OXY	−2.513	0.2736	1.0000	1.0000	0.8852	0.5581
16	APA/FANG	−2.336	0.3558	1.0000	1.0000	1.1004	0.3150
17	DVN/XOM	−2.304	0.3718	1.0000	1.0000	1.9490	0.5623
18	DVN/OXY	−2.273	0.3869	1.0000	1.0000	0.7798	0.5610
19	COP/OXY	−2.203	0.4226	1.0000	1.0000	0.2202	0.9567
20	DVN/EOG	−2.032	0.5120	1.0000	1.0000	1.3795	0.6385
21	EOG/XOM	−2.024	0.5162	1.0000	1.0000	1.2886	0.5895
22	COP/FANG	−1.972	0.5433	1.0000	1.0000	0.3905	0.8980
23	COP/DVN	−1.904	0.5780	1.0000	1.0000	0.2236	0.9153
24	APA/EOG	−1.902	0.5792	1.0000	1.0000	1.4500	0.3635
25	COP/EOG	−1.876	0.5922	1.0000	1.0000	0.5205	0.9637
26	COP/XOM	−1.832	0.6142	1.0000	1.0000	0.5436	0.9697
27	APA/CVX	−1.662	0.6942	1.0000	1.0000	1.0128	0.1810
28	APA/COP	−1.583	0.7279	1.0000	1.0000	0.6140	0.4948

Appendix B. Every trade, gross and net of base-case costs

Direction: +1 is long spread (long EOG, short FANG); −1 is short spread.

#	Dir.	Entry	Exit	Gross P&L	Costs	Net P&L
1	+1	2021-01-05	2021-02-18	−$1,617.94	−$144.11	−$1,762.06
2	−1	2021-05-10	2021-06-16	+$4,037.10	−$131.27	+$3,905.83
3	−1	2021-10-26	2021-11-03	+$2,491.93	−$112.61	+$2,379.31
4	−1	2021-11-10	2021-11-17	+$2,591.44	−$112.05	+$2,479.39
5	+1	2021-12-08	2021-12-14	+$1,793.81	−$108.65	+$1,685.17
6	−1	2022-01-19	2022-02-28	−$1,147.73	−$132.68	−$1,280.41
7	−1	2022-05-09	2022-06-14	+$459.74	−$133.69	+$326.06
8	+1	2022-07-13	2022-08-17	+$1,770.23	−$130.45	+$1,639.78
9	−1	2022-11-08	2022-12-07	+$2,022.30	−$111.63	+$1,910.67
10	+1	2023-02-21	2023-04-04	−$2,574.13	−$124.46	−$2,698.59
11	−1	2023-07-13	2023-08-25	+$837.25	−$132.31	+$704.93
12	+1	2023-09-21	2023-12-07	−$1,478.54	−$134.28	−$1,612.82
13	+1	2024-01-09	2024-04-02	−$5,673.51	−$150.48	−$5,823.98
14	+1	2024-05-30	2024-07-15	+$1,955.74	−$125.59	+$1,830.15
15	−1	2024-08-06	2024-10-23	−$3,007.57	−$145.09	−$3,152.66
16	−1	2024-11-11	2024-12-30	+$1,283.92	−$123.47	+$1,160.44
17	−1	2025-02-07	2025-03-03	−$25.09	−$114.94	−$140.03
18	−1	2025-03-10	2025-03-14	+$3,811.20	−$111.15	+$3,700.05
19	−1	2025-06-23	2025-07-21	+$1,737.63	−$118.90	+$1,618.73
20	+1	2025-09-29	2025-12-31 (forced close)	−$6,014.36	−$140.52	−$6,154.88
			Total	**+$3,253.41**	**−$2,538.32**	**+$715.09**